



Vehicle Battery Range

by GS YUASA



**POWERING HIGH PERFORMANCE
FOR OVER 100 YEARS**

ASIA'S LEADING AUTOMOTIVE BATTERY, NOW AVAILABLE IN EUROPE



Creating the Future of Energy

A GS YUASA GROUP COMPANY

The GS Yuasa Group consists of 65 subsidiaries and 33 affiliates in countries throughout the world.

For over 100 years, the GS Yuasa Group has continually contributed to economic development and the improvement of living standards through the development and manufacture of batteries, power supply systems and lighting equipment.

We are a major force in the market as the world's leading manufacturer of automotive and motorcycle batteries.

Responding to today's increasingly sophisticated needs, our extensive range of next generation energy system lithium-ion batteries encompasses not only vehicle use but also products in a wide range of fields, from deep sea to aerospace.

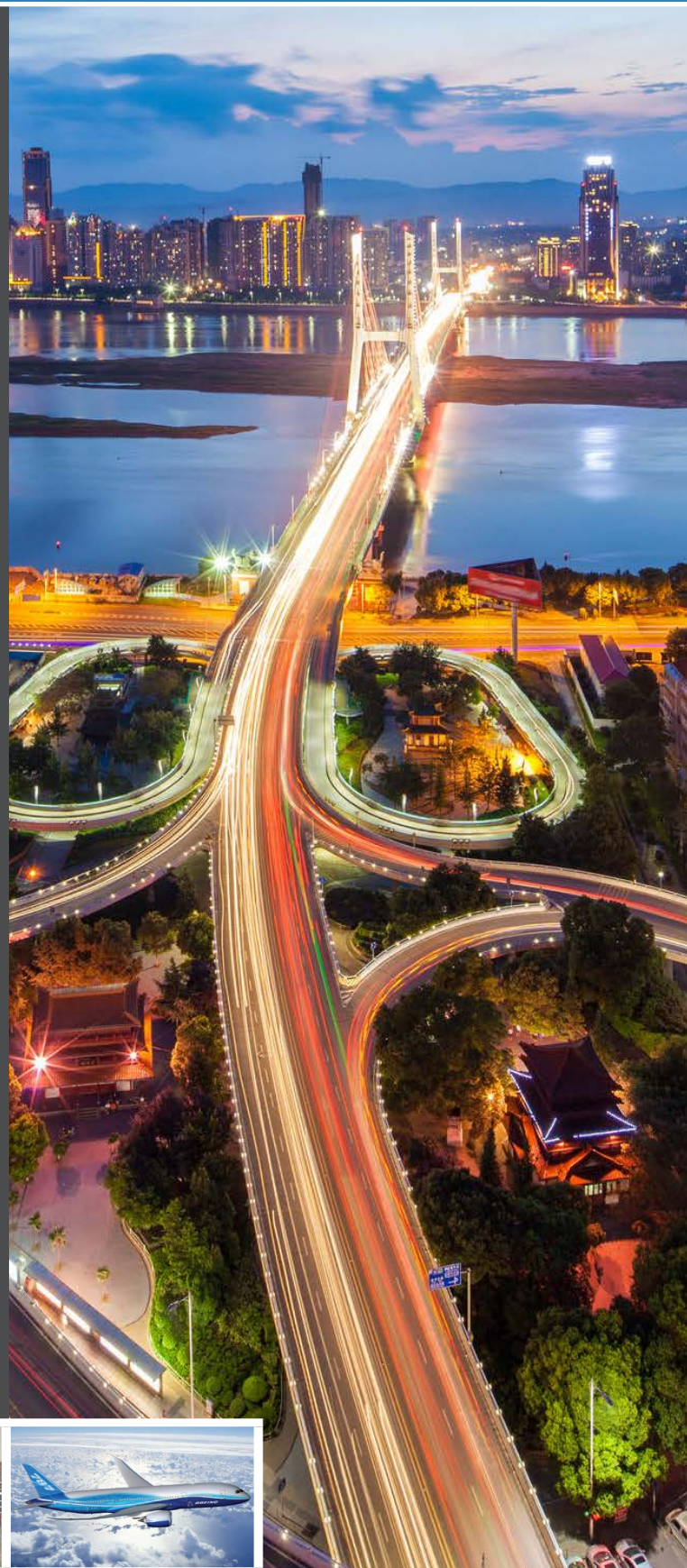
GS YUASA BATTERY EUROPE

For over 30 years, GS Yuasa Battery Europe Ltd have been Europe's leading battery supplier. From sales and distribution centres in Swindon, Milan, Lyon, Madrid, Düsseldorf and Jönköping, GS Yuasa supply European markets with an extensive range of high-quality energy storage and network stabilisation solutions.

Supported by experienced quality assurance, technical support, marketing and customer service teams, our industry leading service and distribution network continues to set new standards in customer care, choice and year-round availability.

Designed to meet the demanding needs of modern vehicles, the new GS automotive range provides superior performance & excellent value.

GS is the leading automotive battery brand in Asia and many other parts of the world. European customers can now enjoy outstanding reliability and power, perfected over a century of GS battery development.



FIND THE RIGHT BATTERY & DETAILED FITTING INFORMATION FAST

With the GS Online Trade Battery Lookup

- Mobile optimised with no login required
- Super-fast number plate, VIN and make/model lookup for automotive, motorcycle & commercial vehicles
- Cross reference search for all DIN, JIS and ETN numbers plus other battery brands & OE references
- Estimated fitting time, battery location & detailed fitting instructions
- Dynamic technical data sheets
- Comprehensive auxiliary and backup battery application information
- Full coverage of the European vehicle parc
- Advanced battery browse function



Used by over 80,000 mechanics and technicians, GS's Online Trade Battery Lookup is Europe's most powerful battery lookup system.

With a host of unique advanced features, it allows users to not only find the right battery, but also access detailed technical and fitting information. It is designed to save users considerable time and money.

Available for all major European markets and languages, the system is free to use online at

batterylookup.gs-battery.com

GS's USB Smart Button provides ultra-fast workshop access to the entire system, just by pushing the button.

YOUR CUSTOMERS CAN FIND THE INFORMATION THEY NEED

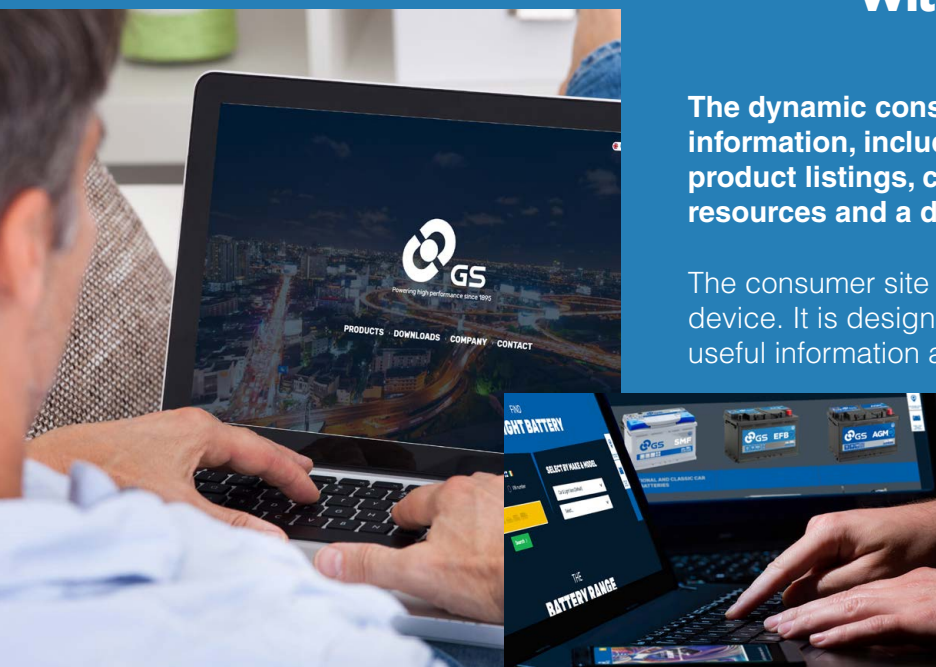
With the GS consumer website

The dynamic consumer GS website is packed with information, including a powerful battery finder, full product listings, company information, downloadable resources and a distributor locator.

The consumer site is mobile optimised so looks great on any device. It is designed to give trade and end users a host of useful information about GS batteries.

Discover more at

www.gs-battery.com



ORIGINAL EQUIPMENT MANUFACTURER

To the world's top vehicle marques



OESAA Member

Vehicle manufacturers demand the best, so overwhelmingly fit GS Yuasa batteries as standard:

Automotive:

BMW, Daihatsu, Daimler, Ford, Fiat, General Motors, Honda, Hyundai, Infiniti, Isuzu, Land Rover, Lexus, Mazda, Mercedes-Benz, Mitsubishi, Nissan, PSA Peugeot Citroen, Renault, Subaru, Suzuki, TATA, Toyota

Motorcycle:

Aprilia, BMW, Ducati, Harley-Davidson, Honda, Kawasaki, KTM, Moto Guzzi, MV Augusta, Piaggio, Suzuki, Triumph, Vespa, Yamaha

Powersport:

Artic Cat, Bombardier Sea-Doo, Bombardier Ski-Doo, BRP Rotax, Honda, Kawasaki, Polaris, Suzuki, Yamaha

Commercial vehicle:

BMC, Daihatsu, Erkunt, Ford Otosan, Hino, Isuzu, GAZ Ticari Araçlar, Hattat Tarim, Honda, Mazda, Mitsubishi Fuso, Mercedes-Benz, Nissan, Otokar, Suzuki, Toyota, Tafe, Temsa, Tumoson, UD Trucks

Plant & agriculture:

Caterpillar, Cummins, IHI Construction Machinery, Iseki, Kobelco, Komatsu, Kubota, Mitsubishi Heavy Industries, Tedano Cranes, Yanmar

Leisure:

Abbey, Ace, Autocruise, Bessacarr, Buccaneer, Compass, Eccles, Elddis, Erwin Hymer Group, Escape, Mondial, Sprite, Sterling, Swift Group, Xplore

GUARANTEED QUALITY

GS Yuasa Battery Sales UK Ltd uses the internationally recognised quality management system ISO 9001 to provide the framework for consistent performance and reliable service in the supply of automotive, motorcycle and industrial batteries to our customers.

Our customers can be confident that we have the tools in place to ensure customer satisfaction in products, supply and service.



FM: 585105



THE GS DIFFERENCE

Your complete power solution



Award-winning products & service

We pride ourselves on offering the best customer service, year-round availability and high quality products. Over recent years, GS Yuasa has been recognised with several prestigious awards including Supplier of the Year from the IAAF.



The largest range available

Providing coverage of over 99% of the European vehicle parc, GS cover more vehicles and applications than any other manufacturer. Our market-leading product ranges include a comprehensive selection of technologically advanced batteries and accessories.



UK based technical support team

With a dedicated team of specialist engineers and technicians, the GS technical department provides support and advice to thousands of GS users each year. The team handle technical enquiries at all levels and can be contacted on +44 (0)1793 833555 or info@gs-yuasa.uk.



Training for workshops & distributors

Training is an integral part of our service. Aiming to dispel myths and highlight opportunities, our interactive sessions deliver up-to-date training on the latest battery technology in an engaging and accessible way.



Innovation & technology

We invest heavily in research, development and product testing to ensure we continue to meet the needs of the market place and the next generation of vehicles.



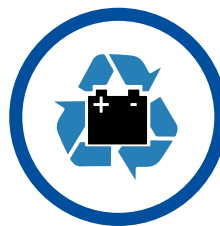
Fast delivery & distribution

Our large distribution centre can deliver mixed pallets or single batteries next day within the UK, and within 48 hours for Northern Ireland and Ireland.



Sponsorship & advertising

GS is one of the most well recognised brands on the market. We're proud of our high-profile sponsorship programme which includes motorsport teams, football, rugby and cricket, plus targeted year-round advertising.



Nationwide scrap collection service

We are dedicated to caring for the environment and offer a comprehensive scrap collection service. Our easy-to-use solution not only collects and recycles waste batteries but also provides businesses with additional revenue.



FOR NEXT GENERATION & START - STOP VEHICLES

Cars & light vans



AGM

START STOP

- Approximately 360,000 starts
- High cyclic durability
- Improved dynamic charge acceptance
- For high specification Start - Stop vehicles
- Absorbed Glass Mat

Product type	Voltage (V)	Capacity (Ah)	CCA (EN1)	Case type	Dimensions (mm)			Layout	Hold-down	Terminal	SOCl	Mean weight (kg)	Performance marking			
					Length	Width	Height						Water Loss	Charge Retention	Vibration	Micro-Cycle
European DIN types																
AGM012	12	50	520	L1 AGM	207	175	190	0	B13	T1		15.9	W5	C2	V3	M3
AGM027	12	60	640	L2 AGM	243	175	190	0	B13	T1		17.7	W5	C2	V3	M3
AGM096	12	70	760	L3 AGM	278	175	190	0	B13	T1		20.5	W5	C2	V3	M3
AGM115	12	80	800	L4 AGM	317	175	190	0	B13	T1		23.2	W5	C2	V3	M3
AGM019	12	95	850	L5 AGM	353	175	190	0	B13	T1		25.6	W5	C2	V3	M3
AGM020	12	105	950	L6 AGM	393	175	190	0	B13	T1		29.6	W5	C2	V3	M3

GS YUASA YU-FIT BATTERY CONFIGURATOR

Battery replacement validation made easy

Most new vehicles have Battery Management Systems (BMS). Failure to reprogram the BMS via the On-Board Diagnostics (OBD) port when a new battery has been fitted can cause serious errors such as loss of the Start - Stop system.

With a robust design and simple operation, the Yu-Fit battery configurator offers great value for money and is one of the most user friendly diagnostic tools on the market. Validate battery replacement quickly and simply with the GS Yuasa Yu-Fit.

Although workshops may already own diagnostic equipment capable of battery validation, to have such a centrally important tool tied up on a relatively straightforward task is not ideal.



FOR NEXT GENERATION & START - STOP VEHICLES

Cars & light vans



EFB START STOP

- Approximately 270,000 starts
- High cyclic durability & dynamic charge acceptance
- For lower specification Start - Stop vehicles
- Sealed tip/tilt double lid:
 - Reduces water loss by up to 30%
 - VDA roll over test compliant
- High charge acceptance type separators
- Anti-sulfation technology
- Carbon/Lithium plate additives
- Integrated flame arrestor

Product type	Voltage (V)	Capacity (A)	CCA (EN)	Case type	Dimensions (mm)			Layout	Hold-down	Terminal	SOCl	Mean weight (kg)	Performance marking			
					Length	Width	Height						Water Loss	Charge Retention	Vibration	Micro-Cycle
EFB012	12	50	540	L1 EFB	207	175	190	0	B13	T1	*	13.4	W3	C2	V3	M1
EFB005	12	64	620	JIS D23/Q85	232	173	225	0	N	T1		16.8	W4	C2	V3	M1
EFB027	12	65	600	L2 EFB	243	175	190	0	B13	T1	*	16.2	W4	C2	V3	M2
EFB100	12	65	650	LB3 EFB	277	175	175	0	B13	T1	*	17.6	W4	C2	V3	M2
EFB096	12	75	700	L3 EFB	278	175	190	0	B13	T1	*	18.8	W4	C2	V3	M2
EFB110	12	75	730	LB4 EFB	315	175	175	0	B13	T1	*	19.8	W4	C2	V3	M2
EFB030	12	80	760	JIS D26/S95	260	173	225	0	N	T1		19.3	W4	C2	V3	M1
EFB335	12	80	780	JIS D31/T110	305	172	225	0	N	T1		22.5	W4	C2	V3	M1
EFB115	12	85	760	L4 EFB	315	175	190	0	B13	T1	*	21.2	W4	C2	V3	M2



AUXILIARY, BACKUP & SPECIALIST

Batteries for next generation cars & light vans



Features

- 9 AGM auxiliary batteries
- 2 Flooded auxiliary batteries
- 3 AGM starter batteries

Fitment examples

- Toyota, Honda & Lexus Hybrids
- Mercedes E & S Class
- Volvo DRIVE
- BMW 7 series
- Mitsubishi & PSA electric vehicles

Product type	Voltage (V)	Capacity (Ah)	CCA (EN1)	Case type	Dimensions (mm)			Layout	Hold-down	Terminal	Mean weight (kg)	Performance marking			
					Length	Width	Height					Water Loss	Charge Retention	Vibration	Endurance
GS Yuasa HJ auxiliary & backup															
HJ-S34B20L	12	35	272	B20 JIS AGM	197	129	227	0	N	T3	10.5	W5	C2	V2	E2
HJ-S34B20L-A	12	35	272	B20 JIS AGM	197	129	227	0	N	T3	10.5	W5	C2	V2	E2
HJ-S34B20R	12	35	272	B20 JIS AGM	197	129	227	1	N	T3	10.5	W5	C2	V2	E2
HJ-S46B24L (S)	12	45	325	B24 JIS AGM	238	126	227	0	N	T1	12.7	W5	C2	V2	E2
HJ-S46B24R	12	45	325	B24 JIS AGM	238	129	227	1	N	T3	12.7	W5	C2	V2	E2
HJ-S55D23L	12	57	430	D23 JIS AGM	220	171	227	0	N	T1	15.1	W5	C2	V2	E2
HJ-S55D23R-B	12	57	430	D23 JIS AGM	220	171	227	1	N	T1	15.1	W5	C2	V2	E2
HJ-S65D26L-B	12	68	400	D26 JIS AGM	257	171	227	0	N	T1	17.3	W5	C2	V2	E2
HJ-S75D31L	12	80	510	D31 JIS AGM	304	171	227	0	N	T1	20.7	W5	C2	V2	E2
Auxiliary & backup															
GTX4L-BS(CP)	12	3.2	50	MC AGM	114	71	86	0	N	5	1.4	-	-	-	-
CB4L-B(CP)	12	4.2	45	MC	120	70	92	0	N	5	1.5	-	-	-	-
GYAUX9	12	8.4	120	MC AGM	151	87	106	1	N	5	3.1	-	-	-	-
GYAUX14	12	12.6	200	MC AGM	150	87	145	1	N	5	4.6	-	-	-	-
GS Yuasa HJ OE JIS Starter															
HJ-A24L	12	40	310	A24 JIS/MX5	238	129	187	0	N	T1	10.5	W5	C2	V2	E2



FOR STANDARD IGNITION VEHICLES

Cars & light vans



SMF

- Approximately 40,000 starts
- Sealed maintenance free lid
- Calcium/Calcium plates
- State of Charge Indicator (SOCi)
- Integrated carrying handle & flame arrestor

Product type	Voltage (V)	Capacity (Ah)	CCA (EN1)	Case type	Dimensions (mm)			Layout	Hold-down	Terminal	SOCl	Mean weight (kg)	Performance marking			
					Length	Width	Height						Water Loss	Charge Retention	Vibration	Endurance
European DIN types																
MF038*	12	35	330	A	243	135	202	1	B12	T1		11.1	W3	C2	V2	E1
SMF202	12	40	330	L0	175	175	190	0	B13	T1		10	W4	C2	V2	E1
SMF077	12	44	360	L1	207	175	190	1	B13	T1	*	12.2	W4	C2	V2	E1
SMF063	12	46	425	LB1	207	175	175	0	B13/14	T1	*	12.2	W4	C2	V2	E1
SMF012	12	52	460	L1	207	175	190	0	B13	T1	*	13	W4	C2	V2	E1
SMF075	12	60	540	LB2	242	175	175	0	B13/14	T1	*	14.6	W4	C2	V2	E1
SMF078	12	60	540	L2	242	175	190	1	B13	T1	*	15.1	W4	C2	V2	E1
SMF027	12	62	540	L2	242	175	190	0	B13/14	T1	*	15.1	W4	C2	V2	E1
SMF100	12	72	700	LB3	278	175	175	0	B13/14	T1	*	17.3	W4	C2	V2	E1
SMF086	12	75	650	L3	278	175	190	1	B13	T1	*	18.3	W4	C2	V3	E1
SMF096	12	75	700	L3	278	175	190	0	B13	T1	*	18.4	W4	C2	V2	E1
SMF110	12	80	740	LB4	315	175	175	0	B13	T1	*	19.3	W4	C2	V2	E1
SMF115	12	82	740	L4	315	175	190	0	B13	T1	*	20.1	W4	C2	V2	E1
SMF017	12	90	760	LB5	353	175	175	0	B13	T1	*	21.3	W4	C2	V2	E1
SMF019	12	95	800	L5	353	175	190	0	B13	T1	*	23.2	W4	C2	V2	E1
SMF020	12	110	900	L6	394	175	190	0	B13	T1	*	26.3	W4	C2	V3	E2
Japanese JIS types																
SMF054	12	40	360	JIS B19	187	127	223	0	N	T1/T3	*	9	W4	C2	V3	E1
SMF055	12	40	360	JIS B19	187	127	223	1	N	T1/T3	*	9	W4	C2	V3	E1
SMF056	12	40	360	JIS NS40	187	135	224	0	B1	T1/T3	*	9.4	W4	C2	V2	E1
SMF053	12	45	400	JIS B24	238	129	224	0	N	T1/T3	*	12.1	W4	C2	V2	E1
SMF057	12	45	400	JIS B24	238	129	224	1	N	T1/T3	*	12.1	W4	C2	V2	E1
SMF108	12	50	420	JIS D20	200	175	225	0	B1	T1	*	12.9	W4	C2	V2	E1
SMF005	12	60	540	JIS D23	232	173	225	0	N	T1	*	15.8	W4	C2	V2	E1
SMF014	12	60	540	JIS D23	232	173	225	1	N	T1	*	15.8	W4	C2	V2	E1
SMF068	12	72	600	JIS D26	264	175	220	0	B07	T1	*	17.6	W4	C2	V2	E1
SMF069	12	72	600	JIS D26	264	175	220	1	B07	T1	*	17.6	W4	C2	V2	E1
SMF334	12	95	800	JIS D31	306	175	222	1	B1	T1	*	22.5	W4	C2	V2	E1
SMF335	12	95	800	JIS D31	306	175	222	0	B1	T1	*	22.5	W4	C2	V2	E1
US BCI group types																
SMF113	12	50	530	US Gp 58	236	182	177	1	B1	T1	*	12.9	W4	C2	V3	E1
Classic European types																
MF048*	12	45	350	E2 (Talbot)	220	135	225	0	B1	T1		13.1	W3	C2	V2	E1
MF049*	12	45	350	E2 (Talbot)	220	135	225	1	B1	T1		13.1	W3	C2	V2	E1
MF072*	12	70	570	D26 T-Lid (Volvo)	258	173	206	1	N	T1		18.8	W3	C2	V2	E1

*Approx. 20,000 starts

FOR CLASSIC & VINTAGE VEHICLES

Cars & light vans



CLASSIC

- For classic & vintage vehicles with 6 Volt electrical systems
- High cyclic durability
- Original equipment (OE) quality, performance & specification
- Authentic period look & feel
- Vibration resistant
- Hard rubber case

Product type	Voltage (V)	Capacity (Ah)	CCA (EN1)	Case type	Dimensions (mm)			Layout	Hold-down	Terminal	SOCl	Mean weight (kg)	Performance marking			
					Length	Width	Height						Water Loss	Charge Retention	Vibration	Endurance
421	6	56	250	Hard Rubber	169	174	217	0	N	T1	-	11.5	W1	C1	V1	E1
404	6	77	360	Hard Rubber	215	171	190	0	B12	T1	-	12.7	W1	C1	V1	E1
501	6	85	385	Hard Rubber	224	173	217	0	N	T1	-	20.1	W1	C1	V1	E1
511	6	105	425	Hard Rubber	224	173	217	0	N	T1	-	23.8	W1	C1	V1	E1
521	6	112	400	Hard Rubber	291	170	227	0	N	T1	-	22.5	W1	C1	V1	E1
531	6	135	630	Hard Rubber	319	170	226	0	N	T1	-	23	W1	C1	V1	E1
541	6	150	510	Hard Rubber	347	174	225	0	N	T1	-	27	W1	C1	V1	E1
451	6	180	700	Hard Rubber	392	175	235	0	N	T1	-	31.2	W1	C1	V1	E1
712	6	200	520	Hard Rubber	315	181	372	0	N	T1	-	39.5	W1	C1	V1	E1
722	6	200	665	Hard Rubber	422	176	255	0	N	T9	-	35	W1	C1	V1	E1
733	6	200	665	Hard Rubber	422	176	240	0	N	T1	-	35	W1	C1	V1	E1
703	6	260	690	Hard Rubber	372	181	372	0	N	T9	-	48	W1	C1	V1	E1
713	6	260	690	Hard Rubber	372	181	372	0	N	T1	-	48	W1	C1	V1	E1

FOR COMMERCIAL VEHICLES

HGV, PSV, specialist utility, agriculture & heavy plant vehicles



EFB

START
STOP

SUPER HEAVY DUTY

- Premium choice for high power demands & end of chassis location
- Next generation enhanced flooded technology suitable for vehicles with Start - Stop systems or standard ignition
- For frequent overnight stays where hotel loads are applied
- XVR Extreme vibration resistance – 8 times V4
- M1 Micro-cycle endurance – over 765 cycles (17.5% DOD)
- Fully maintenance free with sealed double lid & built in electrolyte mixing device to extend battery life
- Optimised for frequent stop & start driving patterns thanks to exceptional partial state-of-charge operation
- OE quality, performance & specification
- Enhanced safety features including integrated flame arrestor & carrying handles

Product type	Voltage (V)	Capacity (Ah)	CCA (EN1)	Case type	Dimensions (mm)			Layout	Hold-down	Terminal	SOCl	Mean weight (kg)	Performance marking			
					Length	Width	Height						Water Loss	Charge Retention	Vibration	Micro-Cycle

European DIN types

EFB629	12	185	1230	DIN B	511	222	215	3	N	T1	*	46.2	W3	C2	V4	M1
EFB625	12	230	1400	DIN C	516	274	236	3	N	T1	*	56.2	W3	C2	V4	M1



FOR COMMERCIAL VEHICLES

HGV, PSV, specialist utility, agriculture & heavy plant vehicles



XVR SUPER HEAVY DUTY

- Designed for high performance vehicles with high power demands
- For regular overnight stays where hotel loads are applied
- XVR Extreme vibration resistance – 8 times V4
- M1 Micro-cycle endurance – over 765 cycles (17.5% DOD)
- Suitable for end of chassis location
- Fully maintenance free with sealed double lid
- End venting for remote exhaust of gases
- Lowest self-discharge rate for extended shelf life
- OE quality, performance & specification
- Enhanced safety features including integrated flame arrestor & carrying handles

Product type	Voltage (V)	Capacity (Ah)	CCA (EN1)	Case type	Dimensions (mm)			Layout	Hold-down	Terminal	SOC1	Mean weight (kg)	Performance marking			
					Length	Width	Height						Water Loss	Charge Retention	Vibration	Micro-Cycle
European DIN types																
XVR627	12	145	900	DIN A	513	189	223	3	N	T1	*	37	W3	C2	V4	M1
XVR623	12	185	1200	DIN B	511	222	215	4	N	T1	*	46	W3	C2	V4	M1
XVR629	12	185	1200	DIN B	511	222	215	3	N	T1	*	46	W3	C2	V4	M1
XVR625	12	230	1350	DIN C	516	274	236	3	N	T1	*	56	W3	C2	V4	M1

Label icons explained



Cycles

Applicable to working out hotel load frequency requirements, cycles are the number of times the battery can be discharged then fully recharged. The higher the cyclic life, the more work the battery can do during its life.



XVR extreme vibration resistance

Batteries with XVR have 8 times V4 vibration resistance. Their advanced plate design and stronger intercell welds mean they are the most robust super heavy duty batteries.



End of chassis location

The battery is built to withstand the extra vibration demands caused by end of chassis location. Failure to use a recommended battery will result in damage and premature failure.



Mixing device

The built-in mixing device constantly mixes the battery electrolyte to prevent plate damage caused by stratification and significantly extends battery service life.

FOR COMMERCIAL VEHICLES

HGV, PSV, specialist utility, agriculture & heavy plant vehicles



SMF SUPER HEAVY DUTY

- Designed for vehicles with standard power demands
- For occasional overnight stays where hotel loads are applied
- Superior vibration resistance – V3
- E2 Endurance rating – up to 150 cycles (50% DOD)
- Fully maintenance free with sealed double lid
- End venting for remote exhaust of gases
- Low self-discharge rate for extended shelf life
- OE quality, performance & specification
- Enhanced safety features including integrated flame arrestor & carrying handles



SMF 623, 625, 627, 628, 629 & 630 batteries feature V4 vibration resistance. They are suitable for end of chassis location.

Product type	Voltage (V)	Capacity (Ah)	CCA (EN1)	Case Type	Dimensions (mm)			Layout	Hold-down	Terminal	SOCl	Mean weight (kg)	Performance marking			
					Length	Width	Height						Water Loss	Charge Retention	Vibration	Endurance
European DIN types																
SMF627	12	143	900	DIN A	513	189	223	3	N	T1	*	36.6	W3	C2	V4	E2
SMF628	12	143	900	DIN A	513	189	223	4	N	T1	*	36.6	W3	C2	V4	E2
SMF630	12	143	900	DIN A	513	189	223	3	B3	T1	*	36.5	W3	C2	V4	E2
SMF623	12	180	1175	DIN B	511	222	215	4	N	T1	*	45.3	W3	C2	V4	E2
SMF629	12	180	1175	DIN B	511	222	215	3	N	T1	*	45.3	W3	C2	V4	E2
SMF625	12	220	1150	DIN C	516	274	236	3	N	T1	*	55.8	W3	C2	V4	E2
European compact types																
SMF663	12	112	870	Euro Compact Low D2	346	173	234	0	N	T1		25.5	W3	C2	V3	E2
SMF664	12	112	870	Euro Compact Low D2	346	173	234	1	N	T1		25.5	W3	C2	V3	E2
SMF665	12	112	870	Euro Compact Low D2A	346	173	234	0	B1	T1		25.5	W3	C2	V3	E2
US commercial types																
SMF640	12	100	800	GR31A BCI	329	173	240	9	N	T1	*	24	W3	C2	V3	E2
SMF641	12	110	925	GR31S BCI	329	173	240	9	N	Thread	*	25	W3	C2	V3	E2
SMF642	12	110	925	GR31A BCI	329	173	240	9	N	T1	*	25	W3	C2	V3	E2
SMF669	12	110	925	GR31 BCI	329	173	240	0	N	T1	*	25	W3	C2	V3	E2



FOR COMMERCIAL VEHICLES

HGV, PSV, specialist utility, agriculture & heavy plant vehicles



MF SUPER HEAVY DUTY

- Designed for vehicles with basic power demands
- High vibration resistance – V2
- E1 Endurance rating – over 80 cycles (50% DOD)
- Maintenance free with easy access lid plugs for topping up if required
- OE quality, performance & specification
- Enhanced safety features including integrated flame arrestor & carrying handles

Product type	Voltage (V)	Capacity (Ah)	CCA (EN1)	Case type	Dimensions (mm)			Layout	Hold-down	Terminal	SOCl	Mean weight (kg)	Performance marking			
					Length	Width	Height						Water Loss	Charge Retention	Vibration	Endurance
European DIN types																
MF643	12	100	680	Euro Compact Low D2	347	174	235	0	N	T1		26	W2	C2	V2	E1
MF644	12	100	680	Euro Compact Low D2	347	174	235	1	N	T1		26	W2	C2	V2	E1
MF663	12	110	750	Euro Compact Low D2	347	174	235	0	N	T1		27.6	W2	C2	V2	E1
MF664	12	110	750	Euro Compact Low D2	347	174	235	1	N	T1		27.6	W2	C2	V2	E1
MF627	12	120	680	DIN A	513	189	223	3	N	T1		33.6	W2	C2	V2	E1
MF655	12	126	750	Euro Compact High D3	347	174	285	0	N	T1		34	W2	C2	V2	E1
MF656	12	126	750	Euro Compact High D3	347	174	285	1	N	T1		34	W2	C2	V2	E1
MF657	12	142	850	Euro Compact High D3	345	175	275	0	N	T1		35.5	W3	C2	V2	E1
MF612	12	143	900	DIN A	513	189	223	3	N	T1		37.6	W2	C2	V2	E1
MF628	12	143	900	DIN A	513	189	223	4	N	T1		37.6	W2	C2	V2	E1
MF630	12	143	900	DIN A	513	189	223	3	B3	T1		37.6	W2	C2	V2	E1
MF621	12	155	900	DIN B	513	223	223	3	N	T1		42	W2	C2	V2	E1
MF620	12	180	1100	DIN B	513	223	223	3	B3	T1		44.9	W2	C2	V2	E1
MF623	12	180	1100	DIN B	513	223	223	4	N	T1		44.9	W2	C2	V2	E1
MF626	12	180	1100	DIN B	513	223	223	4	B3	T1		44.9	W2	C2	V2	E1
MF629	12	180	1100	DIN B	513	223	223	3	N	T1		44.9	W2	C2	V2	E1
MF624	12	200	1100	DIN C	513	274	242	4	N	T1		55	W2	C2	V2	E1
MF625	12	200	1100	DIN C	513	274	242	3	N	T1		55	W2	C2	V2	E1
MF632	12	220	1150	DIN C	513	274	242	3	N	T1		57	W2	C2	V2	E1
Japanese JIS types																
MF616	12	110	680	E41 JIS	407	176	234	0	N	T1		29.6	W2	C2	V2	E1
MF618	12	110	680	E41 JIS	407	176	234	1	N	T1		29.6	W2	C2	V2	E1
US commercial types																
MF222	12	135	900	MAC 143	510	216	210	3	N	T1		38	W2	C2	V2	E1
MF615	12	135	910	MAC 110	514	175	210	3	B3	T1	*	32.5	W3	C2	V3	E1
MF622	12	150	900	MAC 143	510	216	210	3	N	T1		39.2	W2	C2	V2	E1
Other types																
MF633	12	140	900	Volvo	360	253	240	6	B1	T1		40.8	W2	C2	V2	E1
MF638	12	140	1100	Tractor	508	175	205	0	B1	T1		36.5	W2	C2	V2	E1

FOR FLOOR CLEANING & AERIAL ACCESS



PRO-SPEC

- Deep cycle performance
- Extended cyclic design life of over 500 cycles
- Resistant to vibration
- Low maintenance with easy access vent caps for topping up
- Glass mat/porous rubber separator construction to resist corrosion & reduce electrical resistance
- Special case & lid designed to prevent short circuit
- Suitable applications, including:
 - Industrial vehicles
 - Mobility vehicles
 - Warehouse equipment
 - Fork lifts
 - Access platforms
 - Floor cleaners

Part number	Capacity				Dimensions (mm)				Weight	Terminal
	@25A (Mins)	@75A (Mins)	5HR (Ah)	20HR (Ah)	Length	Width	Height	Terminal height	(kg)	
Deep cycle 6 Volt										
DCB 605-6	383	105	175	210	259	179	245	276	27	DT / ET
DCB 105-6	447	115	185	225	259	179	245	276	28.6	DT / ET
DCB 125-6	488	132	195	240	259	179	245	276	30.7	DT / ET
DCB 145-6	530	145	215	260	259	179	264	295	33	DT / ET
Deep cycle 8 Volt										
DCB 875-8	295	75	145	170	262	181	245	276	29	DT / ET
DCB 890-8	340	90	155	190	262	181	245	276	31.6	DT / ET
DCB 8125-8	425	110	190	240	262	181	283	316	37.6	DT / ET
Deep cycle 12 Volt										
DCB 1275-12	290	70	125	150	329	181	245	276	37.5	ET

Terminal

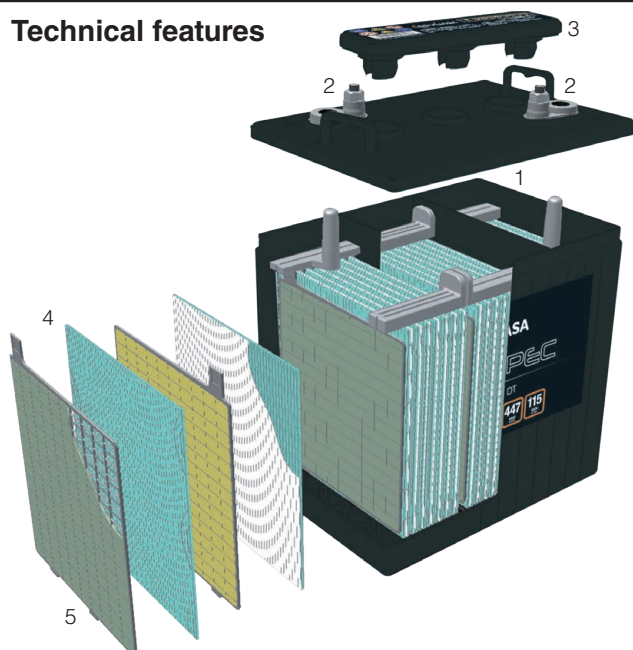
DT Dual Fit Terminal



ET Embedded Terminal



Technical features



- Case/Lid:
 - Less weight, shock resistant & acid resistant thanks to PP Resin construction
 - Specially designed structure to prevent short circuit from active material shedding in the bottom
- Terminal:
 - Cast with special lead alloy
 - Special plating to minimize heat generation electric resistance
 - Designed to be vibration resistance
 - Easily detachable with standard & bolt/nut structure
- Cap:
 - Designed & engineered to allow gas venting
 - Easy to refill & maintain
- Separator:
 - Porous rubber material protects against acid corrosion
 - Excellent physical characteristics and lower electric resistance
 - Uses micro-fibre glass mat to minimise active material shedding
- Plates:
 - Negative - 99.9% pure lead with hard paste feature, specific additives for excellent deep cycle ability
 - Positive - corrosion resistant grid with hard paste feature, specific additives for deep cycle purpose

FOR CARAVANS, MOTORHOMES & SMALL BOATS



LEISURE & MARINE

- Optimised for repeated charging & discharging
- Original equipment quality & specification
- Fully maintenance free
- Supplied charged & ready to fit
- Enhanced safety features
- Built to last

GS Leisure EFB and AGM utilise the latest in next generation battery technology to provide greater endurance, more available cycles and faster charging.

AGM also offers maximum safety with spill proof internal construction.

Product Type	Voltage (V)	Capacity (A)	CCA (EN1)	MCA A	Watt Hours	Cycles	NCC Class	Dimensions (mm)			Weight (kg)	Features				
								Length	Width	Height		Case	Layout	Terminal	Hold-down	Lid
L26-70	12	70	480	580	840	80	C	269	174	225	17	GR24 BCI	1	T1	B5	MF
L26-80	12	80	560	680	960	80	C	269	174	225	18	GR24 BCI	1	T1	B5	MF
L35-90	12	90	680	820	1080	80	C	352	175	227	23.6	EU Lo D2	1	T1	N	MF
L35-100	12	100	720	875	1200	90	C	352	175	227	24.7	EU Lo D2	1	T1	N	MF
L36-100	12	100	900	1000	1200	160	C	353	175	190	23.3	L5 DIN	0	T1	B13	SMF
L35-115	12	115	750	915	1380	80	C	352	175	227	25.4	EU Lo D2	1	T1	N	MF
L28-EFB	12	75	730	900	900	200	-	278	175	190	19.3	L3 DIN	0	T1	B13	SMF
L31-EFB	12	100	750	900	1200	220	B	330	173	240	26	GR31 BCI	0	TM	N	SMF
L36-EFB	12	100	850	1050	1200	200	B	353	175	190	24	L5 DIN	0	T1	B13	SMF
L26-AGM	12	90	300	350	1080	420	A	259	168	236	27	GR24 BCI	1	M6(F)/T1	N	VRLA
L36-AGM	12	95	850	1050	1140	400	A	353	175	190	25.6	L5 DIN	0	T1	B13	VRLA
Dual purpose leisure and marine																
DLMA-145	12	145	900	1080	1740	-	-	513	189	223	37	DIN A	3	T1	N	SMF
DLMB-185	12	185	1200	1440	2220	-	-	513	223	223	46	DIN B	3	T1	N	SMF
DLMC-230	12	230	1350	1620	2760	-	-	513	274	242	56.7	DIN C	3	T1	N	SMF

FOR GARDEN & SPECIALIST APPLICATIONS



GARDEN & SPECIALIST

- Rugged construction resists vibration & deep discharge damage
- L-type terminal allows quick & easy installation
- Fully maintenance free
- Supplied charged & ready to fit
- Enhanced safety features including integrated flame arrestor
- For ride on mowers, kit cars & specialist

FOR MOTOR BOATS, SAILING BOATS, YACHTS & WATERCRAFT



MARINE START

- For marine starting applications
- Vibrolock construction provides enhanced vibration resistance
- Dual terminal
- Supplied charged & ready to fit
- Enhanced safety features
- Fully maintenance free
- Leak proof double lid

MARINE DUAL EFB

GS Marine Dual EFB batteries are developed to meet the demands of both starting and auxiliary power requirements. They utilise next generation technology to provide greater endurance, more available cycles and faster charging.



Product Type	Voltage (V)	Capacity (A)	CCA (EN1)	MCA A	Watt Hours	Cycles	NCC Class	Dimensions (mm)			Weight (kg)	Features				
								Length	Width	Height		Case	Layout	Terminal	Hold-down	Lid

Marine Start

M26-80S	12	80	600	720	960	-	-	260	174	225	18.9	GR24 BCI	1	TM	B5	SMF
M27-90S	12	90	720	860	1080	-	-	304	173	225	21.2	GR27 BCI	1	TM	B1	SMF
M31-100S	12	100	750	900	1200	-	-	330	173	240	23.5	GR31 BCI	1	TM	N	SMF

Marine Dual EFB

M26-EFB	12	80	600	780	960	-	-	260	174	225	19.7	GR24 BCI	1	TM	B5	SMF
M27-EFB	12	90	700	840	1080	-	-	304	173	225	22.3	GR27 BCI	1	TM	B1	SMF
M31-EFB	12	100	750	900	1200	-	-	330	173	240	26	GR31 BCI	1	TM	N	SMF

Product type	Voltage (V)	Capacity (Ah)	CCA (EN1)	Dimensions (mm)			Weight (kg)	Layout	Terminal	SMF
				Length	Width	Height				
895	12	26	250	187	127	181	7.5	0	T4	
896	12	26	250	187	127	181	7.5	1	T4	
U1R	12	30	330	194	126	183	7.8	0	T4	*
U1	12	30	330	194	126	183	7.8	1	T4	*
897	16	2.2	330	Husqvarna Robotic Mower Battery 5351209-01						



FOR MOTORCYCLE & POWERSPORT

Motorcycles, scooters, ATVs, UTVs, jet skis & snowmobiles

GS high performance maintenance free motorcycle and powersport batteries feature:

- Fully maintenance free – never add water, but periodic charging is required when not in use
- Premium AGM (Absorbed Glass Mat) technology with no free acid
- Sealed, spill-proof design with multi-angle fitment
- Genuine high-performance OE battery developed to meet motorcycle manufacturers' custom requirements
- Sulphate Stop additive to protect against undercharging damage
- Supplied filled & ready to fit



YTZ HIGH PERFORMANCE MAINTENANCE FREE AGM

These compact, lightweight AGM VRLA batteries deliver greater starting power from a more volume efficient unit.

- Maximum power from minimum space - up to 30% more CCA than standard AGM batteries despite being lighter & more compact
- Can be mounted at extreme angles with no risk of leakage
- Ideal for sport bikes & also motorcycles with a Start - Stop system
- Brass terminals for improved conductivity & optimised performance
- Exceptional resistance to vibration & corrosion
- Constructed with high purity lead



GTX & GT HIGH PERFORMANCE MAINTENANCE FREE AGM

A range of high performance AGM VRLA batteries, incorporating extra battery plates with a specialist design for increased cranking amps, less maintenance and longer life.

- Increased cranking power - up to 30% more CCA than standard AGM batteries
- High resistance to vibration & corrosion
- Up to four times longer life than competitors' equivalent product



FOR MOTORCYCLE & POWERSPORT

Motorcycles, scooters, ATVs, UTVs, jet skis & snowmobiles

GS motorcycle and powersport batteries feature:

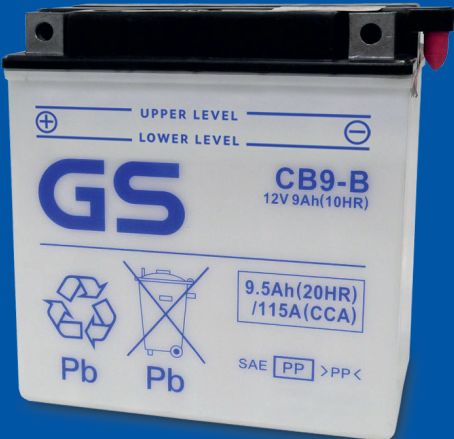
- Up to four times longer life than competitors' equivalent product
- Genuine high-performance OE battery developed to meet motorcycle manufacturers' custom requirements
- Sulphate Stop additive to protect against undercharging damage



GTX & GT MAINTENANCE FREE AGM

Ideal for motorcycles, scooters, ATVs, ride on mowers and personal watercraft, these permanently sealed batteries do not require topping up. Aside from periodic charging, they remain maintenance free for the life of the battery.

- Increased cranking power - up to 30% more CCA than conventional type batteries
- Superior resistance to vibration & corrosion
- Fully maintenance free once activated
- Ideal for sport bikes & also motorcycles with a start - stop system
- Premium AGM technology with no free acid
- Sealed & spill-proof design



CB CONVENTIONAL

CB batteries are designed to meet the special demands of personal watercraft, snowmobiles and ATVs. Dependable power, reduced maintenance and longer life plus a unique power boosting design makes the range ideal for touring bikes and modified vehicles.

- High cranking power - constructed with special thin plate separators
- Low maintenance, water loss & self-discharge with easy access vent plug design
- Heavy-duty glass mat technology for high vibration resistance
- Heat sealed case to protect against seepage & corrosion



12V CONVENTIONAL

A rugged, dependable battery engineered to withstand vibration and deliver reliable cranking power.

- High cranking power - constructed with special thin plate polyethylene separators
- Engineered to protect against corrosion & vibration damage
- Easy access vent plug design for maintenance
- Up to four times longer life than competitors' equivalent product
- Premium AGM technology with no free acid
- Sealed & spill-proof design

UNDERSTANDING BATTERY RATINGS

Batteries are rated according to a range of specifications and standards. Understanding these ratings and their relevance to the battery's application and operating conditions is key to selecting the right battery.

CCA (Cold Cranking Amperes)

This is a measurement of the maximum current a fully charged battery can deliver at -18°C. It is used to determine a battery's ability to supply high cranking current to start the vehicle's engine and maintain sufficient voltage to power the ignition requirements under severe cold starting conditions. Temperature plays a key role in a battery's ability to deliver CCA.

700
A (EN)

Ampere-Hour @ 20hr (Ah)

Ah refers to the battery's storage capacity. At 25°C, the battery must achieve greater than 20 hours of discharge time at a given load, down to a cut-off voltage of 10.5V. For example, a 60Ah battery will deliver a current of 3A for 20 hours. The more accessories that are in use, the faster available battery energy is consumed. Consumption of the battery's capacity is referred to as 'discharging' the battery.

75
Ah

New Battery Performance Markings

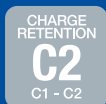
Alongside the common voltage (V), capacity (Ah) and cold cranking amps (A) ratings, all GS automotive batteries carry recently introduced EN50342-1/6 European Standard performance markings.

Every battery in the range has undergone extensive testing to determine its performance. Having passed the requirements of several stringent tests, each battery has earned a classification number for the following criteria:



Water Loss: rated W1 – W5

The ability to retain water content. The better a battery performs, the less water it will lose in service.
W1 (Traditional lead-acid battery) - W5 (Premium next generation battery technology)



Charge Retention: rated C1 - C2

The ability of the battery to retain stored chemical energy (charge) when not in use.
C1 (Traditional lead-acid) - C2 (Modern calcium)



Vibration Level: rated V1 – V4

The battery's physical ability to resist the potentially damaging effects of vibration.
V1 (Traditional car and light van) - V4 (Extreme heavy duty plant and commercial vehicle)



Endurance: rated E1 – E4 (Conventional types only)

The ability to withstand repeated charge/discharge cycles to 50% depth of discharge without failure.
E1 (Conventional car and light van) - E4 (Extreme heavy duty plant and commercial vehicle)



Micro-Cycle: rated M1 – M3 (EFB & AGM types only)

The ability of the battery to provide power to restart the engine after frequent stop phases, its ability to recover state of charge afterwards and the aging effects due to shallow pulse loads.
M1 (Basic start-stop) - M3 (High end start-stop and hybrid)

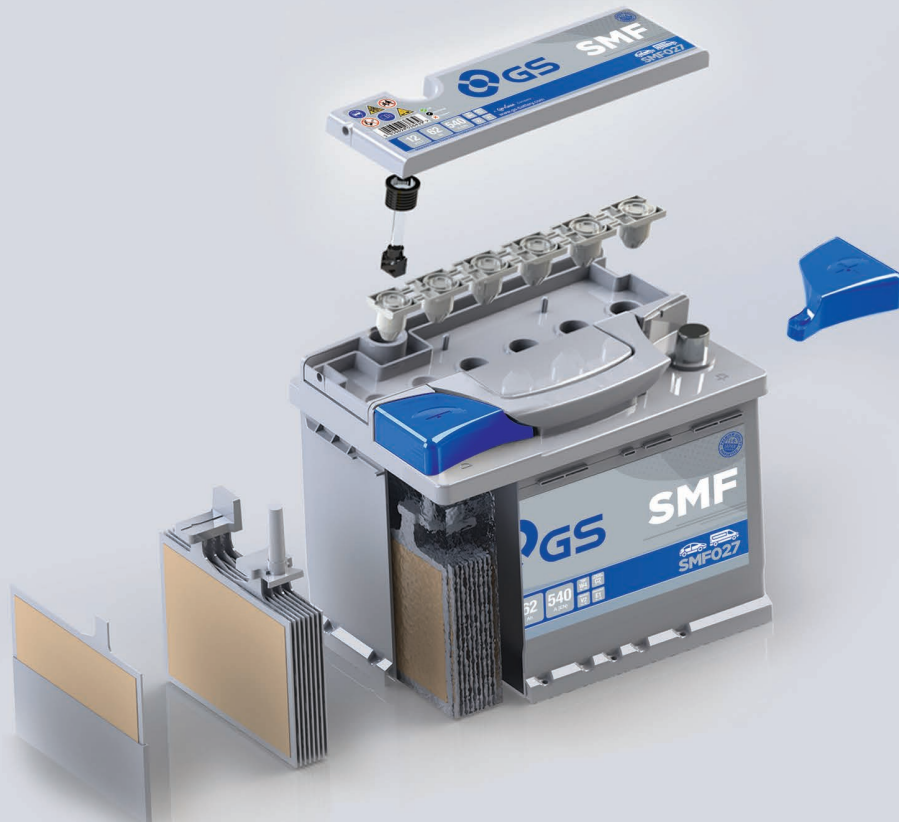
GUARANTEED QUALITY

GS Yuasa Battery Sales UK Ltd uses the internationally recognised quality management system ISO 9001 to provide the framework for consistent performance and reliable service in the supply of automotive, motorcycle and industrial batteries to our customers.

GS Yuasa customers can be confident that we have the tools in place to ensure customer satisfaction in products, supply and service.



FM: 585105



OTHER RATINGS

1200
WATT HOURS

Watt Hours

The amount of power the battery can supply when fully charged. You can calculate your requirements by working out the consumption of the accessories the battery will power, and how long you usually use them for.

UP TO
160
CYCLES

Cyclic Life

The number of times the battery can be discharged to 50% then fully recharged. The higher the cyclic life, the more work the battery can do during its life in service.

960
MCA

Marine Cranking Amps (MCA)

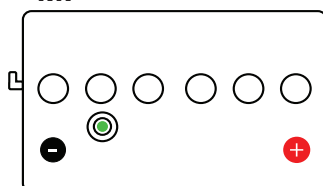
The amount of cranking power the battery can provide to start the engine. It is the number of Amperes it can deliver for 30 seconds. Unlike CCA / A (EN), MCA is measured at a temperature of 0°C as this is more relevant to marine use.

Reserve Capacity (RC)

The length of time in minutes that a fully charged battery at 25°C can deliver a current of 25 Amps until the voltage drops to 10.50V (5.25V for a 6V Battery). RC provides an indication of the time that a vehicle with a normal electrical load will run for if the charging system (alternator) fails.

Leisure & marine lid types explained

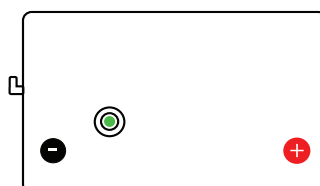
MF



Maintenance Free

End vented - must be installed with breather pipe in occupied spaces.

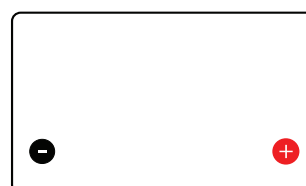
SMF



Sealed Maintenance Free

Rollover proof sealed lid. End vented - must be installed with breather pipe in occupied spaces.

VRLA

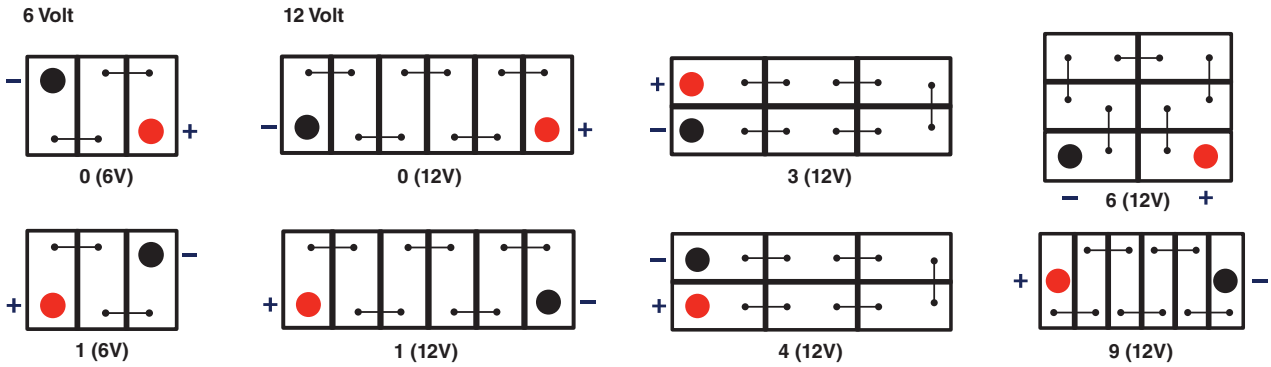


Valve Regulated Lead Acid

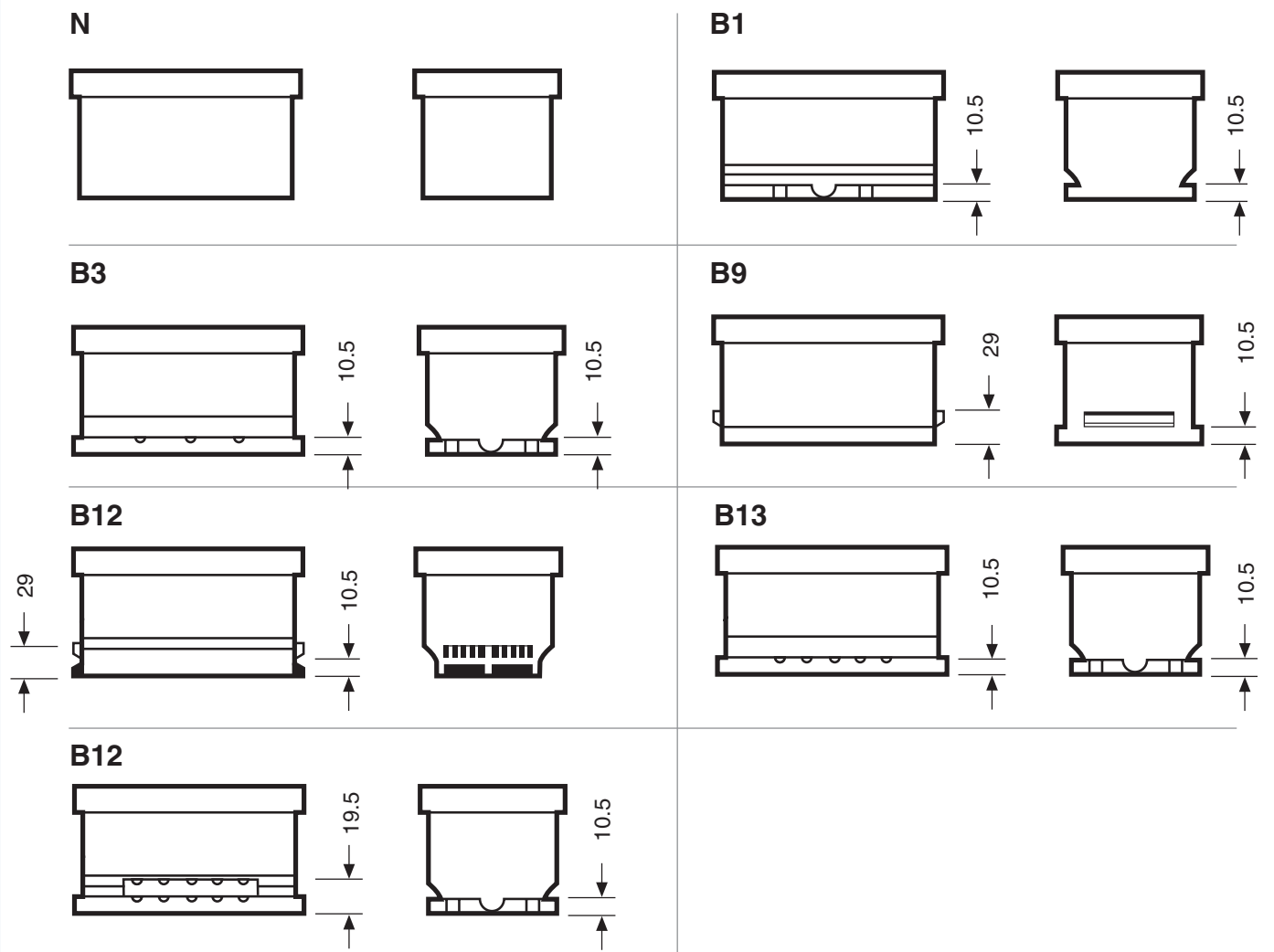
Sealed, spill proof construction with no free acid. Can be installed in occupied spaces with natural ventilation.

All types are maintenance free so do not require topping up with water during normal service.

CELL ASSEMBLY LAYOUTS



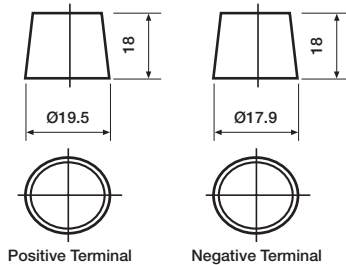
HOLD DOWN & CONTAINER FEATURES



TERMINAL CONFIGURATIONS

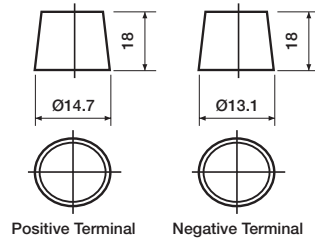
T1

Standard DIN post



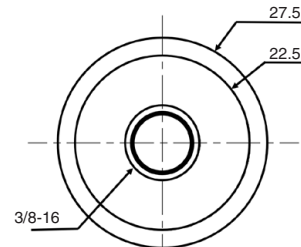
T3

Small Japanese JIS terminal

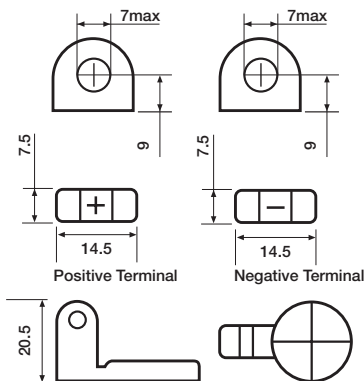


FT

US BCI group type



T4

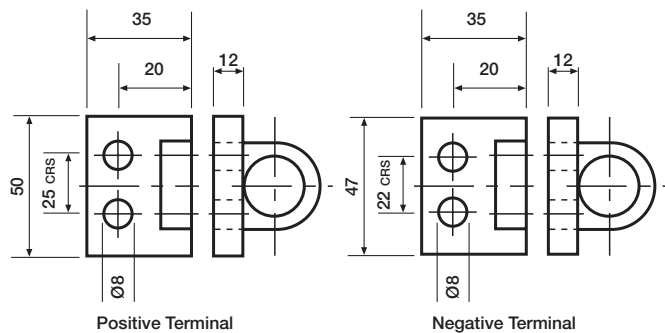


5



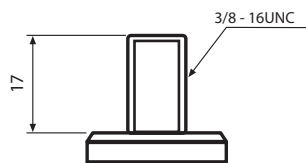
T9

Classic 702/703/721/722



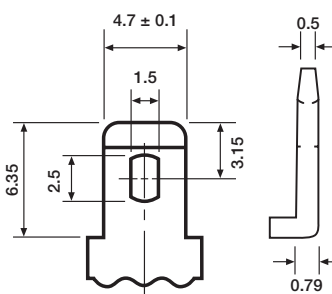
Thread

US BCI stud terminal



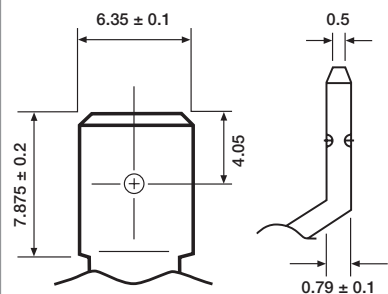
A

Faston tab:187



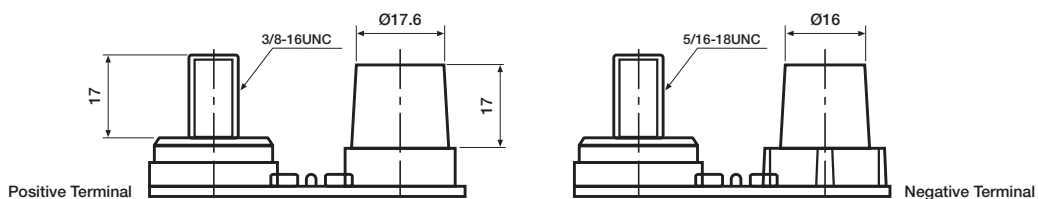
A/D

Faston tab:250



TM

Marine dual terminals



(Dimensions in millimetres)

ASIA'S LEADING AUTOMOTIVE BATTERY NOW AVAILABLE IN EUROPE



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Consumer website: www.gs-battery.com

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Cat. No.	LIT059	10/25
E. & O. E.		